THE MENINGES & DURAL VENOUS SINUSES

LEARNING OBJECTIVES

At the end of the lecture, the student should be able to :

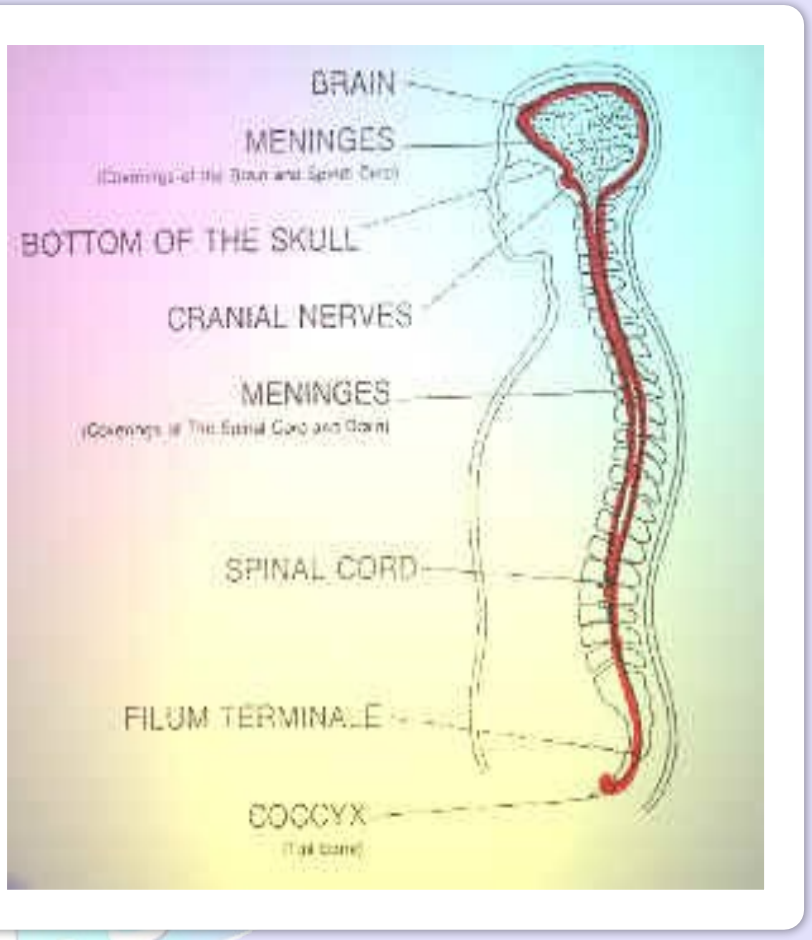
- Define Meninges.
- Know Functions of Meninges.
- Know Composition of Meninges.
- Describe Dura Mater.
- Describe the Dural infoldings.
- Explain the Dural Venous Sinuses.
- Describe the Arachnoid and the Pia Mater.
- Describe the Meningeal Spaces.

THE MENINGES

Definition:

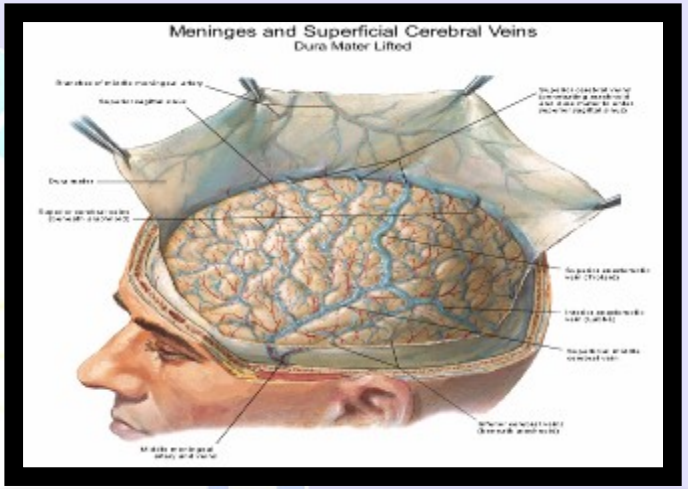
Membranous coverings of the brain that lie immediately internal to the cranium.

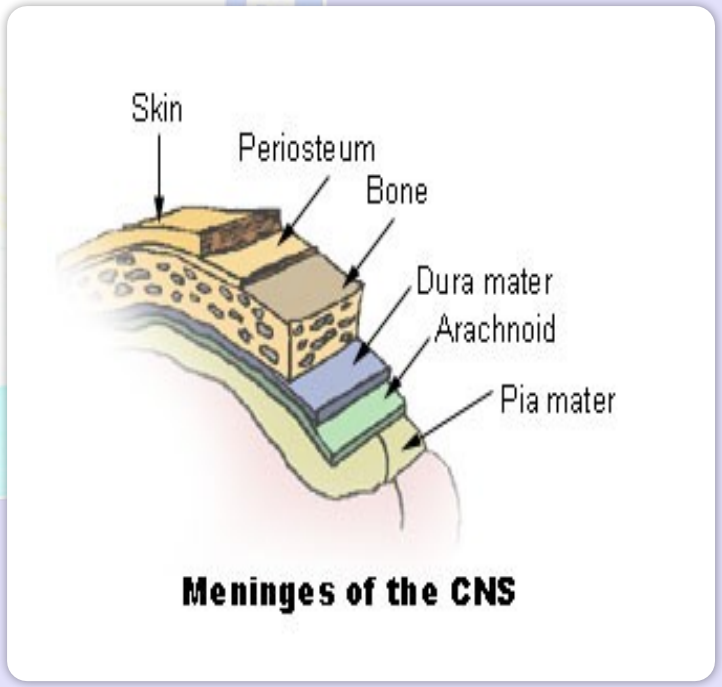
- **The cranial meninges:**
  - Protect the brain.
  - Form the supporting framework for arteries, veins and venous sinuses.
  - Enclose a fluid-filled cavity, the subarachnoid space, which is vital to the normal function of the brain.



COMPOSITION OF MENINGES

- **The meninges are composed of three membranous connective tissue layers:**
  - **Dura Mater :**
    - tough, thick external fibrous layer.
  - **Arachnoid Mater(arachnoid):**
    - thin intermediate layer.
  - **Pia Mater (pia):**
    - delicate internal vascularized layer.





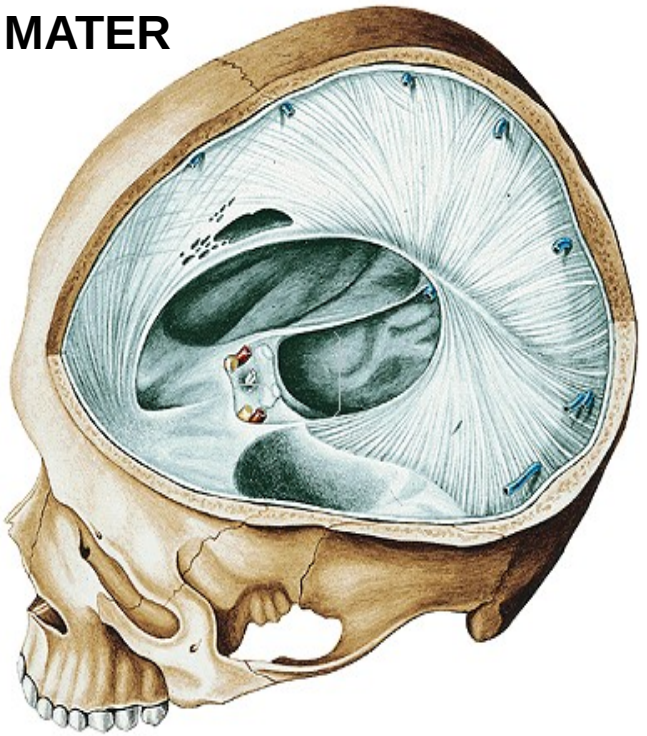
DURA MATER

- **DURA MATER** is a thick, durable membrane, closest to the skull.

- It has two layers
  - The endosteal layer
  - The meningeal layer.

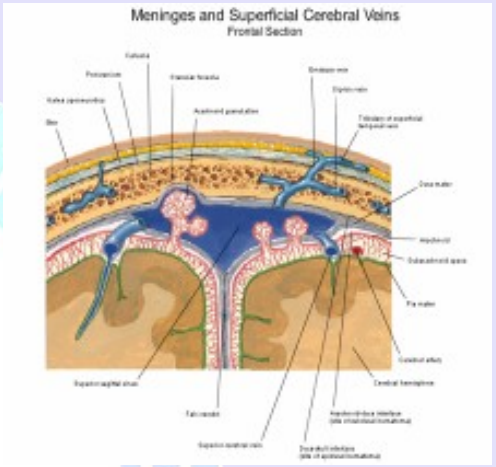
- **Endosteal layer** is the periosteum covering the inside of the skull.
- **Meningeal layer** is dense, strong, fibrous layer covering the brain
- Through the foramen magnum it is continuous with the dura mater of the spinal cord.

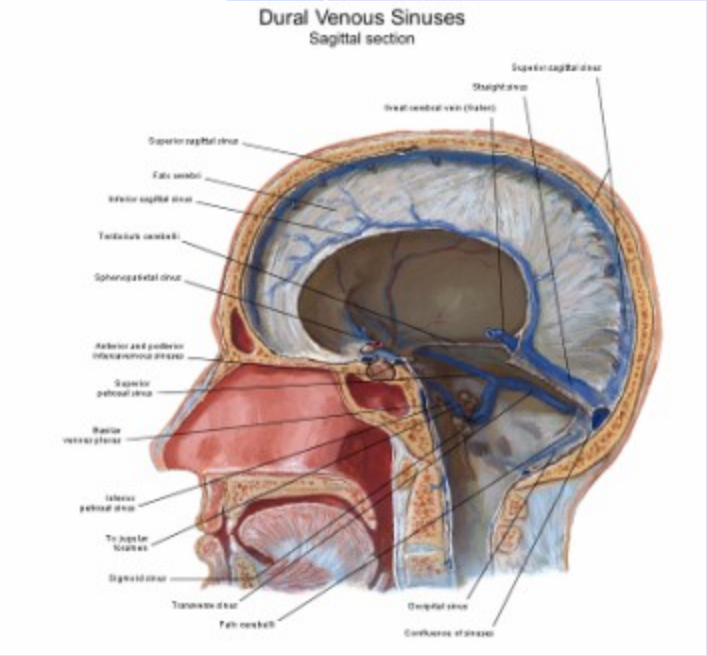
DURA MATER



FALX CEREBRI

- Sickle shaped fold of dura mater which lies in the midline in the longitudinal fissure between the two cerebral hemispheres
- Anteriorly attached to the internal frontal crest and the crista galli of ethmoid.
- Broad posterior end fuses with the upper surface of the tentorium cerebelli

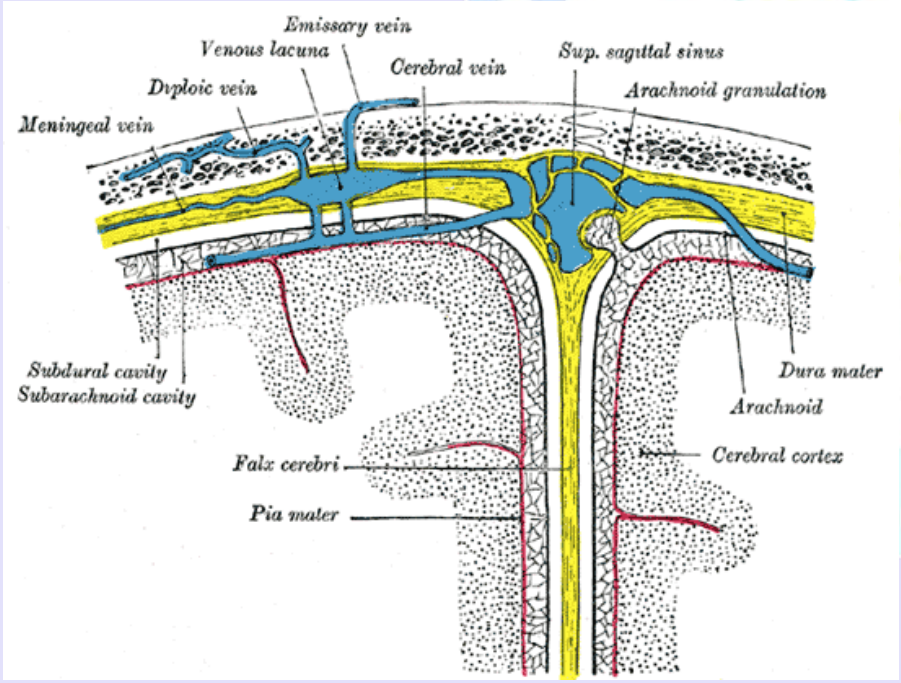


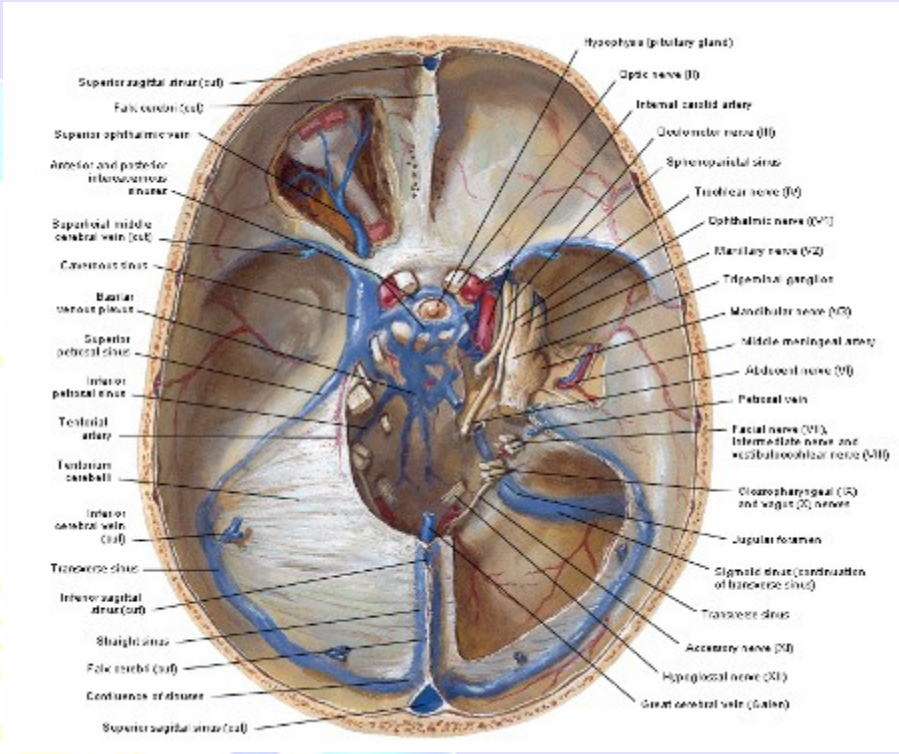


Superior sagittal sinus runs in its upper attached margin.

Inferior sagittal sinus runs in its lower free margin.

Straight sinus runs in its attachment to the tentorium cerebelli.

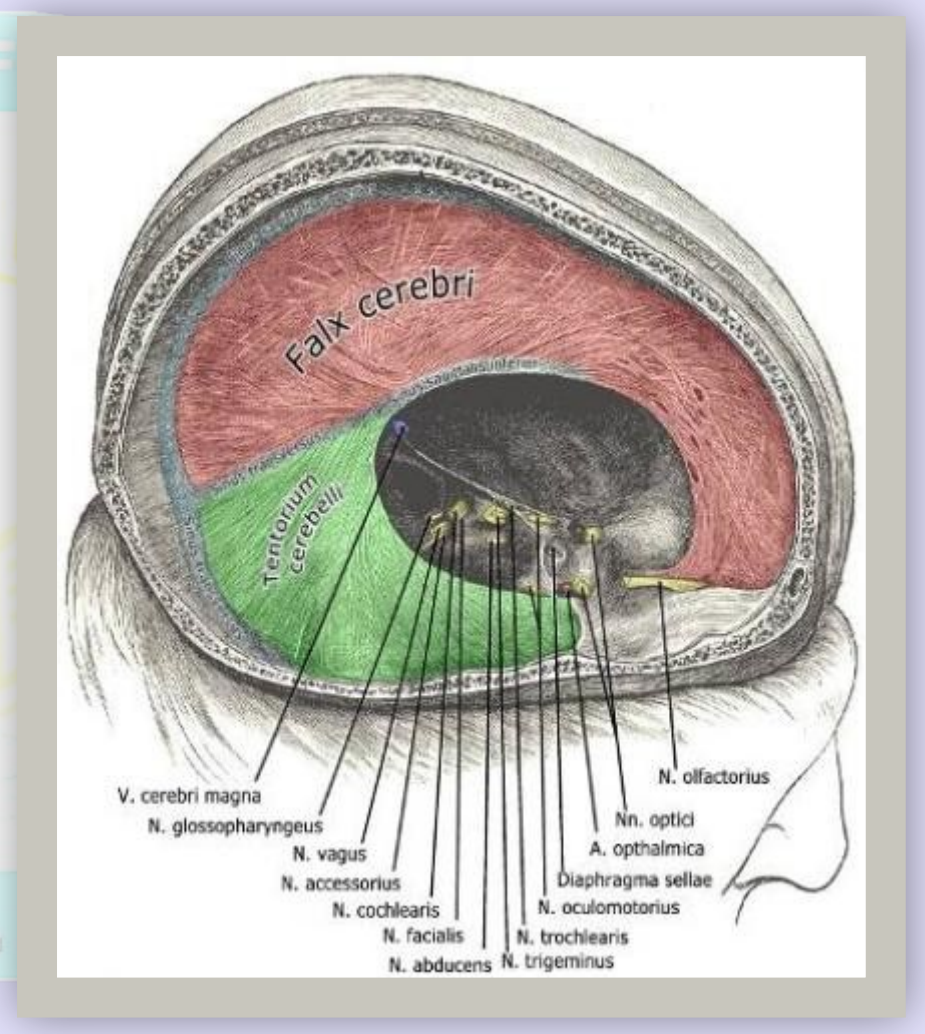




Diagrammatic representation of a section across the top of the skull

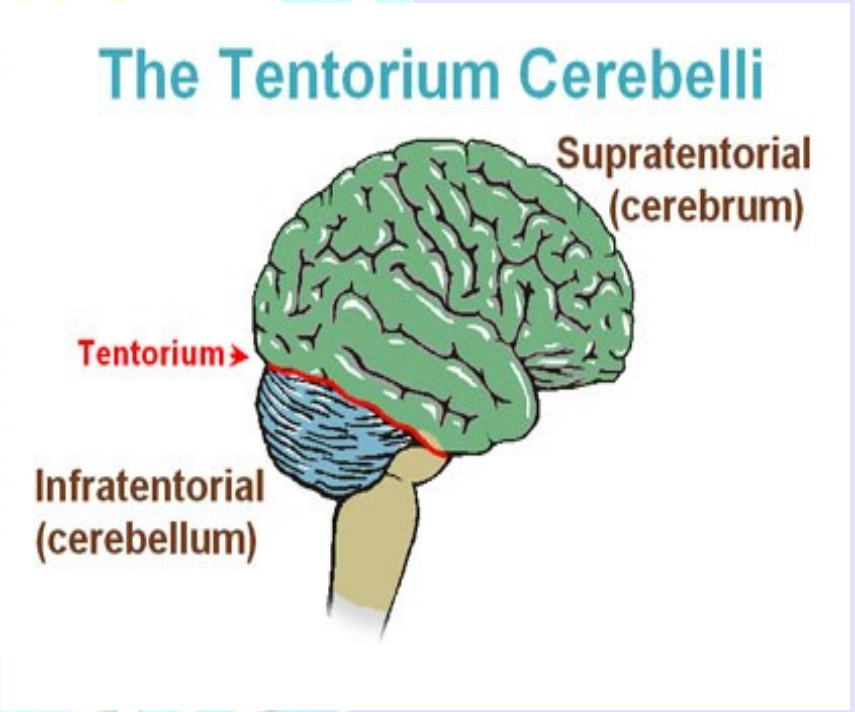
THE TENTORIUM CEREBELLI

- Crescent shaped fold of dura mater that roofs the posterior cranial fossa.
- Has a tentorial notch at its anterior free margin for the passage of midbrain.
- Recess for the trigeminal ganglion and nerve is the forward extension of the lower layer of tentorium cerebelli near the apex of the petrous temporal bone.



TENTORIUM CEREBELLI

- It covers the superior surface of the cerebellum, and supports the occipital lobes of the brain.
- Anterior border is free and concave, and bounds a large oval opening for the transmission of the cerebral peduncles.
- Attached, behind, by its convex border, to the transverse ridges upon the inner surface of the occipital bone, and there encloses the transverse sinuses; in front, to the superior angle of the petrous part of the temporal bone on either side, enclosing the superior petrosal sinuses.
- To the middle line of its upper surface the posterior border of the falx cerebri is attached, the straight sinus being placed at their line of junction.



CLINICAL SIGNIFICANCE

- Tentorium is important because brain tumors are often characterized as supratentorial (above the tentorium) and infratentorial (below the tentorium).
- Location of tumor can help in determining the type of tumor.
- Most childhood tumors are infratentorial, while most adult tumors are supratentorial.
- Since the tentorium is a hard structure, if there is a volume expansion in the parenchyme above the tentorium, the brain can get pushed down partly through the tentorium.
- This is called herniation and will often give mydriasis on the affected side, due to pressure on cranial nerve III (n. Oculomotor).
- Tentorial herniation is a serious symptom, especially since the brainstem is likely to be compressed as well if the intracranial pressure rises further.

THE FALX CEREBELLI

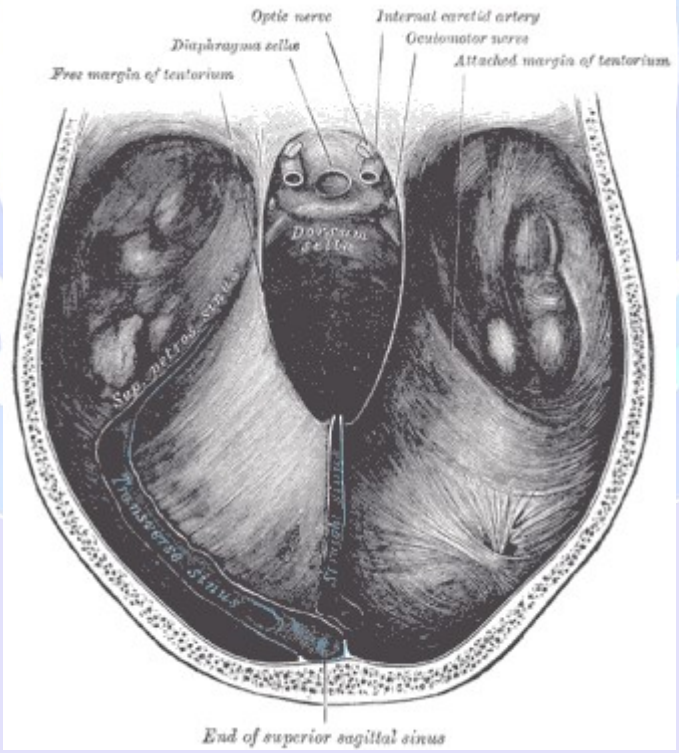
- is a small sickle shaped fold of dura mater attached to the **internal occipital crest, which projects between the two** cerebellar hemispheres received into the Posterior cerebellar notch .

- Its posterior free margin contains the occipital sinus.

- As it descends, it sometimes divides into two smaller folds, which are lost on the sides of the foramen magnum.

DURA MATER CONTD..

- The diapragma sellae or sellar diaphragm is the circular fold of dura mater that almost completely roofs the fossa hypophyseos in the sphenoid bone of the skull or it forms the roof of the sella turcica.
- It retains the pituitary gland in the fossa hypophyseos, with only the infundibulum of the pituitary gland passing through it.
- It has a central opening for the stalk of the hypophysis.



Diaphragma sellae

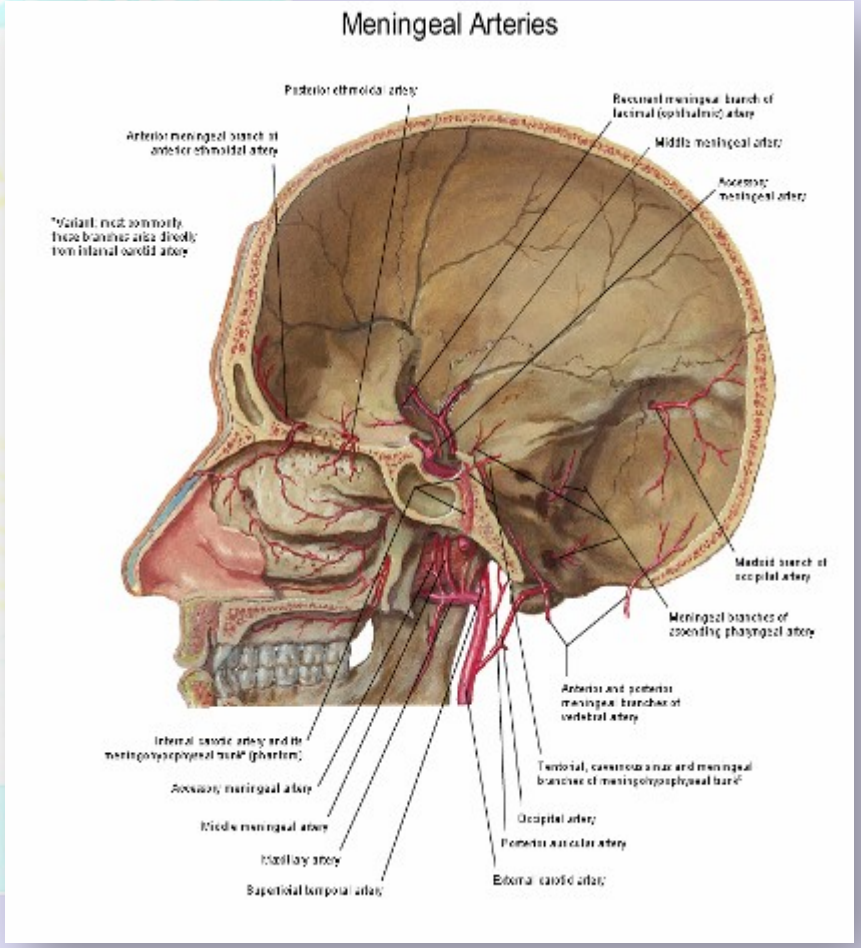
DURAL NERVE SUPPLY

- Trigeminal
- Vagus
- Upper three cervical nerves
- Branches from the sympathetic trunk

- Numerous nerve endings are sensitive to stretching

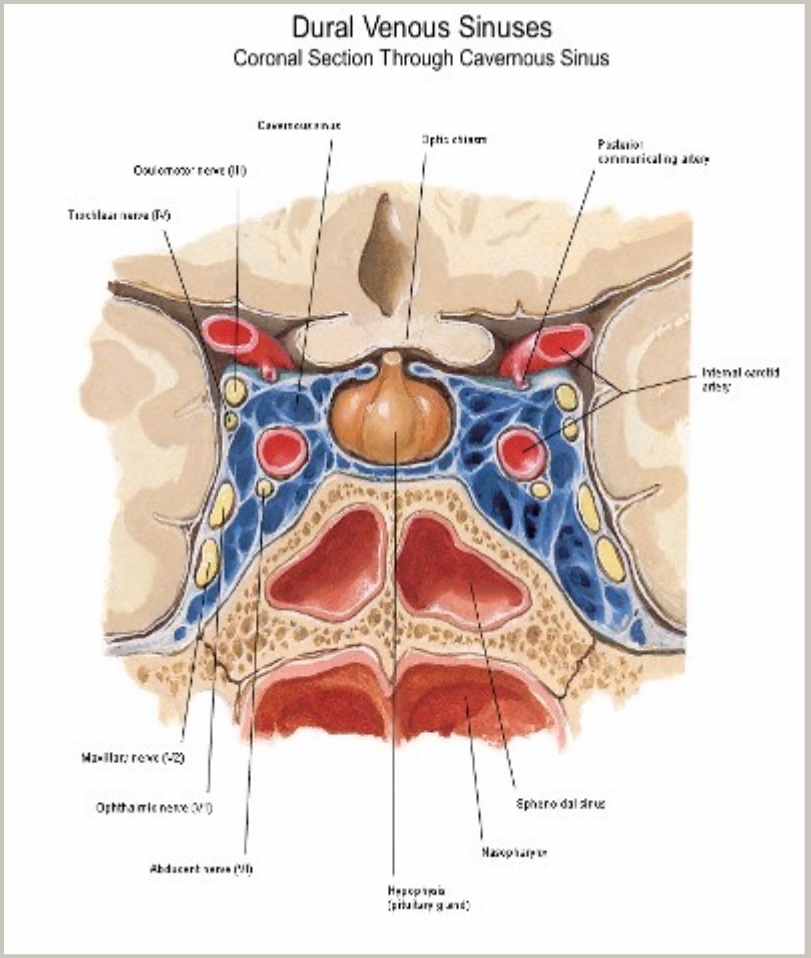
DURAL ARTERIAL SUPPLY

- Branches from the
  - Internal carotid
  - Maxillary
  - Ascending pharyngeal
  - Occipital
  - Vertebral arteries
  - Middle meningeal is clinically the most important artery



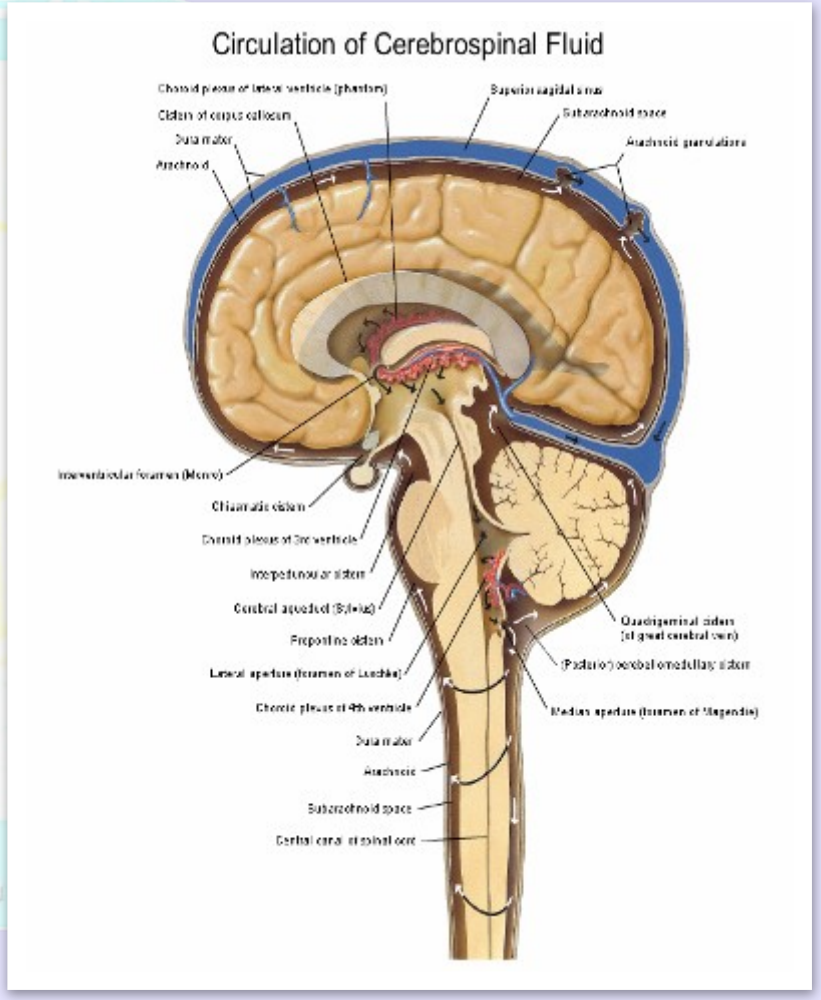
DURAL VENOUS SINUSES

- Situated between the layers of dura mater
- receive blood from the brain through the cerebral veins and CSF from the subarachnoid space through the arachnoid villi
- Blood ultimately drains into the internal jugular veins
- Lined by endothelium
- Thick walls, no valves, no muscular tissue
- Connected to the diploic veins and the veins of the scalp through the valveless emissary veins



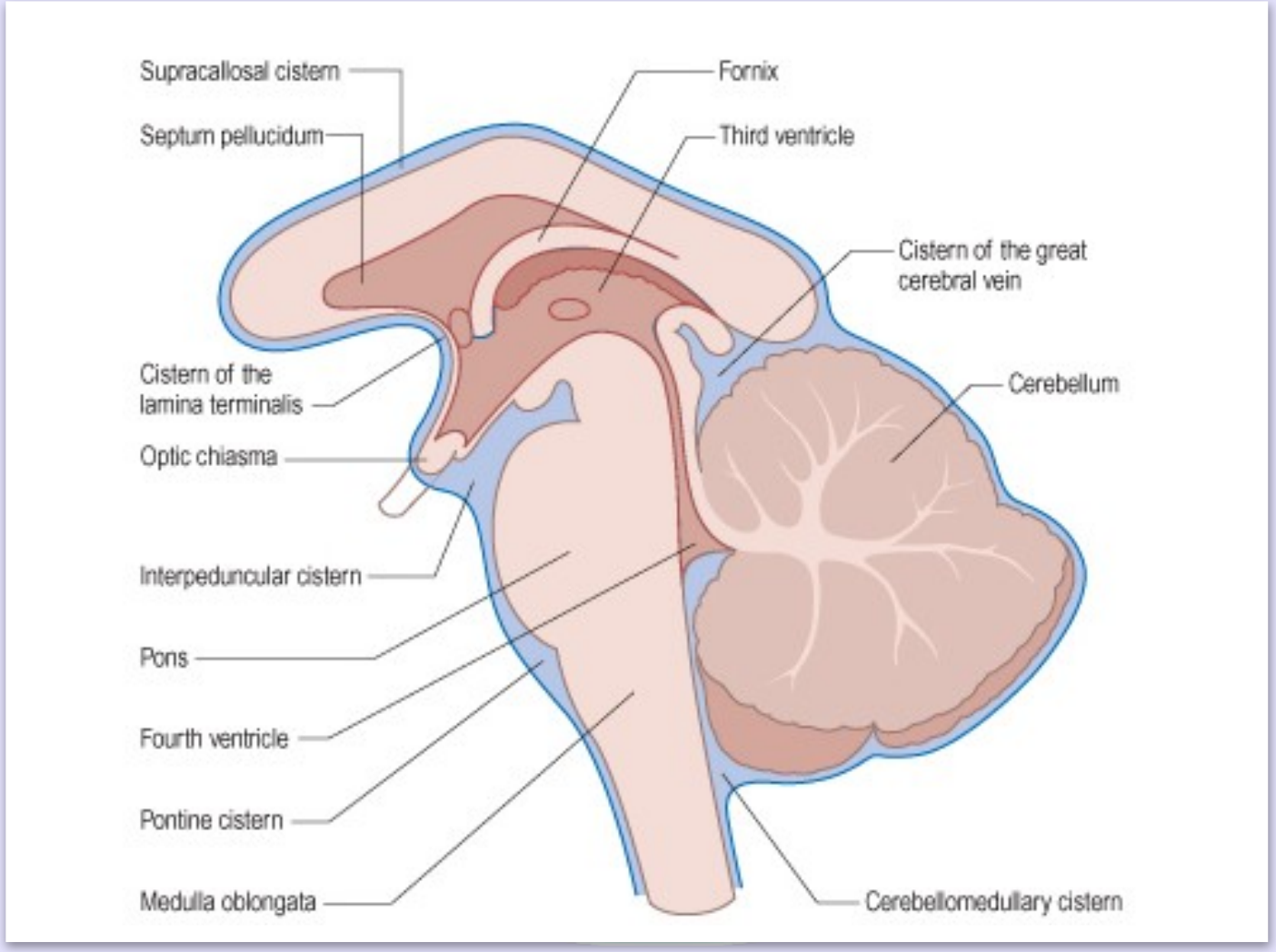
Dural venous sinuses contd..

- **The superior sagittal sinus** occupies the upper fixed border of falx cerebri
- Begins at the foramen cecum, grooves the vault of the skull, and the internal occipital crest deviates usually to the right to become continuous with the transverse sinus
- Receives cerebrospinal fluid from the venous lacunae and blood from the superior cerebral veins
- Confluence of sinuses is the dilated posterior part of the sinus



Dural venous sinuses contd..

- **The inferior sagittal sinus** occupies the lower free margin of the falx cerebri
- It joins the great cerebral vein at the free margin of tentorium cerebelli and forms the straight sinus
- Receives few cerebral veins from the medial aspect of the cerebral hemispheres
- **The straight sinus** occupies the line of junction between the falx cerebri and the tentorium cerebelli
- It ends by turning usually to the left to form the transverse sinus


Supracallosal cistern
Septum pellucidum
Fornix
Third ventricle
Cistern of the great cerebral vein
Cerebellum
Cistern of the lamina terminalis
Optic chiasma
Interpeduncular cistern
Pons
Fourth ventricle
Pontine cistern
Medulla oblongata
Cerebellomedullary cistern

Dural venous sinuses contd..

- **The transverse sinuses** begin at the internal occipital protuberance and occupy the attached margins of tentorium cerebelli
- They receive the superior petrosal sinuses, the inferior cerebral and cerebellar veins and the diploic veins
- They end by turning downward as the sigmoid sinuses
- **The sigmoid sinuses** are related to the mastoid antrum on either side
- They open into the superior bulb of the internal jugular vein on either side
- The occipital sinus is present in the attached margin of the falx cerebelli
- It is connected below to the vertebral veins and above to the confluence of sinuses

Dural venous sinuses contd..

- **The cavernous sinus** is situated in the middle cranial fossa on each side of the body of sphenoid bone
- Crossed by numerous trabeculae
- Extends from the superior orbital fissure anteriorly to the apex of the petrous temporal bone behind
- The internal carotid artery with its sympathetic plexus and the abducent nerve pass through the sinus
- Third and fourth cranial nerves and the ophthalmic and maxillary divisions of trigeminal nerve occupy the lateral wall of the sinus

Dural venous sinuses contd..

- Tributaries of cavernous sinus are the superior and inferior ophthalmic veins, the sphenoparietal sinus and the central vein of the retina
- Drainage is to the superior and inferior petrosal sinuses behind and the pterygoid venous plexuses below
- Anterior and posterior intercavernous sinuses unite the two cavernous sinuses anterior and posterior to the hypophyseal stalk

DURAL SPACES

- The subarachnoid space is the space which normally exists between the arachnoid and the pia mater, which is filled with cerebrospinal fluid.
- Normally, the dura mater is attached to the skull, or to the bones of the vertebral canal in the spinal cord. The arachnoid is attached to the dura mater, and the pia mater is attached to the central nervous system tissue.
- When the dura mater and the arachnoid separate through injury or illness, the space between them is the subdural space.

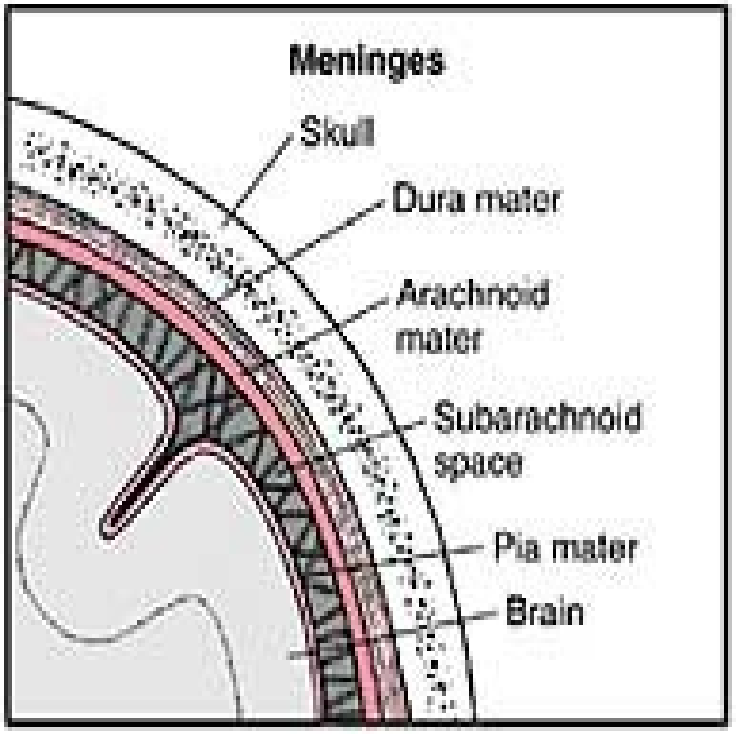


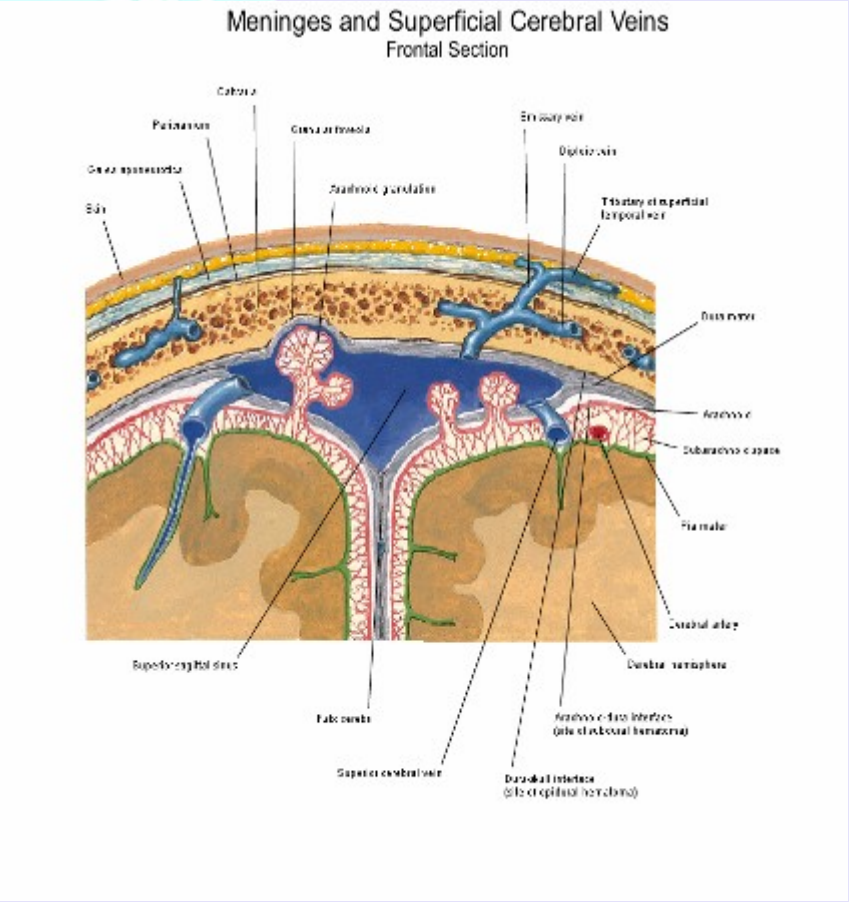
Figure 2

PATHOLOGY

- There are three types of hemorrhage involving the meninges
- A subarachnoid hemorrhage is acute bleeding under the arachnoid; it may occur spontaneously or as a result of trauma.
- A subdural hematoma is a hematoma (collection of blood) located in a separation of the arachnoid from the dura mater. The small veins which connect the dura mater and the arachnoid are torn, usually during an accident, and blood can leak into this area.
- An epidural hematoma similarly may arise after an accident or spontaneously.
- Other medical conditions which affect the meninges include meningitis (usually from fungal, bacterial, or viral infection) and meningiomas arising from the meninges or from tumors formed elsewhere in the body which metastasize to the meninges.

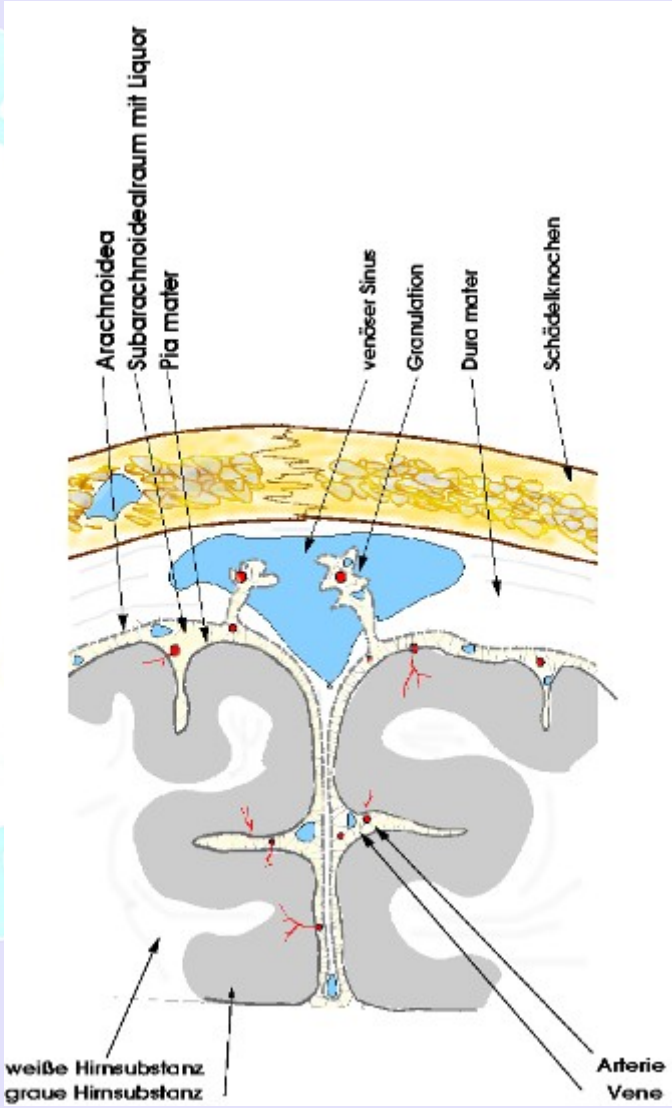
ARACHNOID MATER

- The arachnoid and pia mater are sometimes considered, as a single structure the *leptomeninx*, (Lepto- from the root meaning *thin* in Greek).
- It is interposed between the two other meninges, the more superficial dura mater and the deeper pia mater, and is separated from the pia mater by the subarachnoid space.
- Delicate impermeable membrane lying between the dura and pia mater
- Subarachonoid space is the space between the arachnoid and pia mater, filled with cerebrospinal fluid
- The membrane bridges over the sulci of the brain.



ARACHNOID MATER CONTD..

- Projects into the venous sinuses to form the arachnoid villi
- Most numerous along the superior sagittal sinus
- Aggregates are called arachnoid granulations
- Cerebrospinal fluid diffuses into the venous sinuses through the arachnoid villi
- The arachnoid fuses with the epineurium of the nerves at their point of exit from the skull
- The subarachnoid space extends into the orbital cavity along the optic nerve
- Cerebrospinal fluid is produced by the choroid plexuses of the lateral, third and fourth ventricles



ARACHNOID GRANULATION

- **Arachnoid granulations** (or **arachnoid villi**) are small protrusions of the arachnoid (the thin second layer covering the brain) through the dura (the thick outer layer).

- They protrude into the venous sinuses of the brain, and allow cerebrospinal fluid (CSF) to exit the brain, and enter the blood stream.

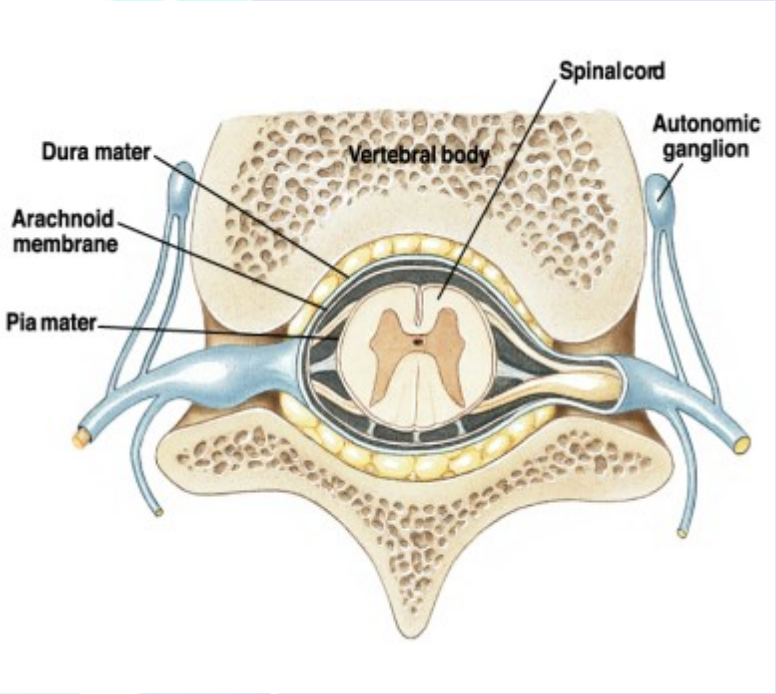
- Largest granulations lie along the superior sagittal sinus, a large venous space running from front to back along the centre of the head (on the inside of the skull).

ARACHNOID MATER CONTD..

- Subarachnoid cisternae are the spaces where pia and arachnoid are widely separated
  - Cisterna cerebellomedullaris
  - Cisterna interpeduncularis
  - Chiasmatic cistern
  - Prepontine cistern
  - Cistern of great cerebral vein

PIA MATER OF THE BRAIN

- The **pia mater** (Latin: "tender mother") is the delicate innermost layer of the meninges—the membranes surrounding the brain and spinal cord.
- Vascular fold covered by flat mesothelial cells
- Closely invests the brain
- Dips into the deepest sulci
- Covers the cerebral arteries entering the brain
- Fuses with the epineurium of the cranial nerves
- Forms the tela choroidea of the roof of the third and fourth ventricles of the brain, and fuses with the ependyma to form the choroid plexuses in the lateral, third and fourth ventricles of the brain



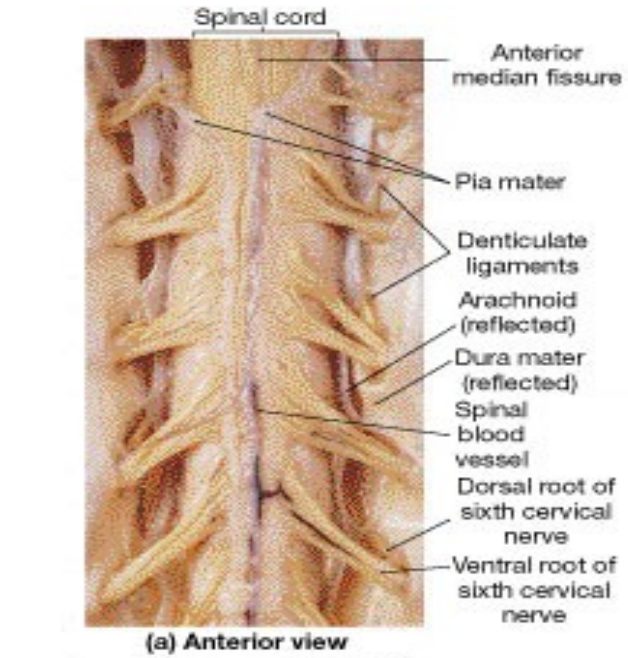
PIA MATER OF THE BRAIN

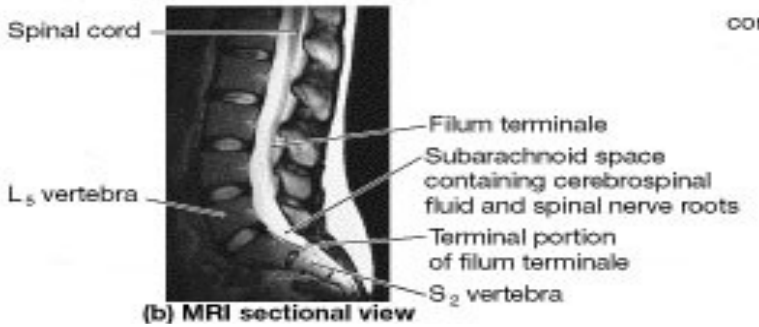
- The thin, mesh-like pia mater closely envelops the entire surface of the brain, running down into the fissures of the cortex.
- It joins with the ependyma which lines the ventricles to form choroid plexuses that produce cerebrospinal fluid.
- In the spinal cord, the Pia mater attaches to the Dura mater by the denticular ligaments through the arachnoid membrane.
- The pia mater is a neural crest derivative.

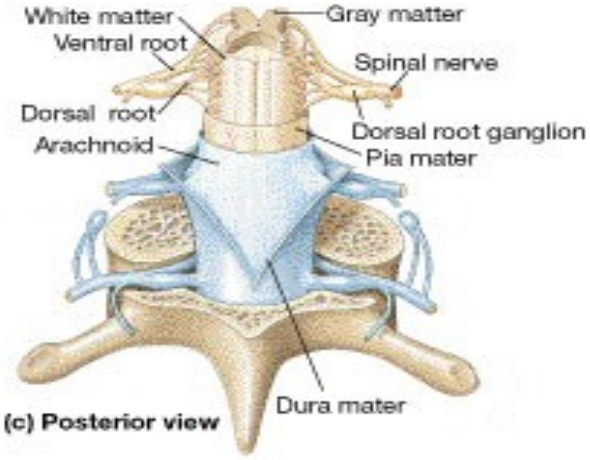
COVERINGS OF THE SPINAL CORD

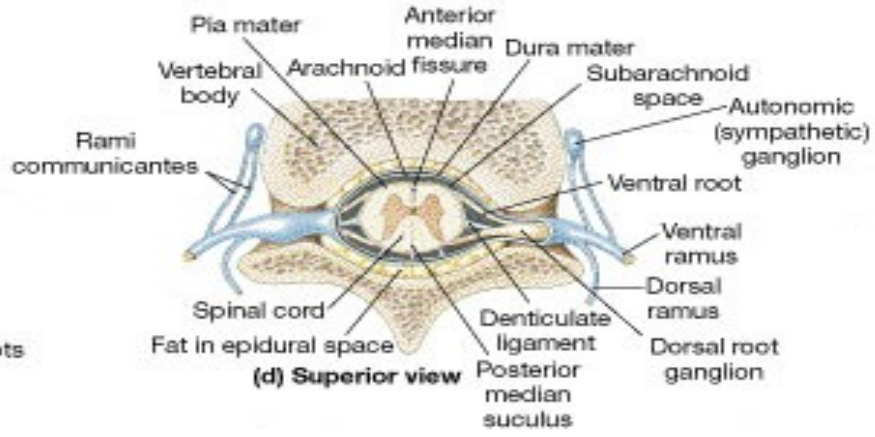
- 3 layers called meninges
- Dura mater
  - Outer layer
  - Dense irregular connective tissue
- Arachnoid
  - Middle layer
  - Encloses cerebrospinal fluid
- Pia mater
  - Adheres tightly to the surface of the spinal cord

MENINGES OF THE SPINAL CORD





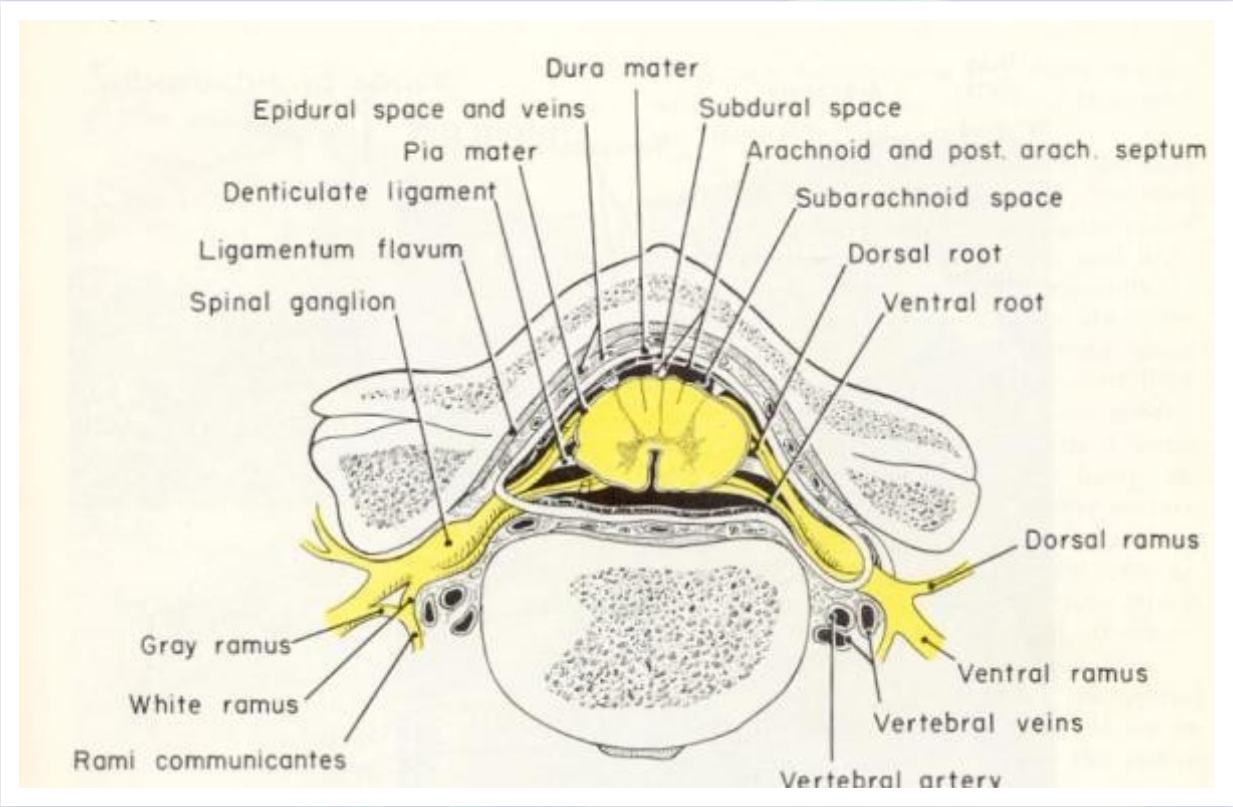




SPACES BETWEEN THE MENINGES

- ***Epidural Space***—a space between the dura mater and the wall of the vertebral canal; contains fat and connective tissue
- ***Subdural Space***—a thin space between the dura mater and the arachnoid mater; contains interstitial fluid
- ***Subarachnoid Space***—between the arachnoid mater and the pia mater; contains ***cerebrospinal fluid***

The dura mater of the spinal cord



Periosteum

Epidural space

Dura mater

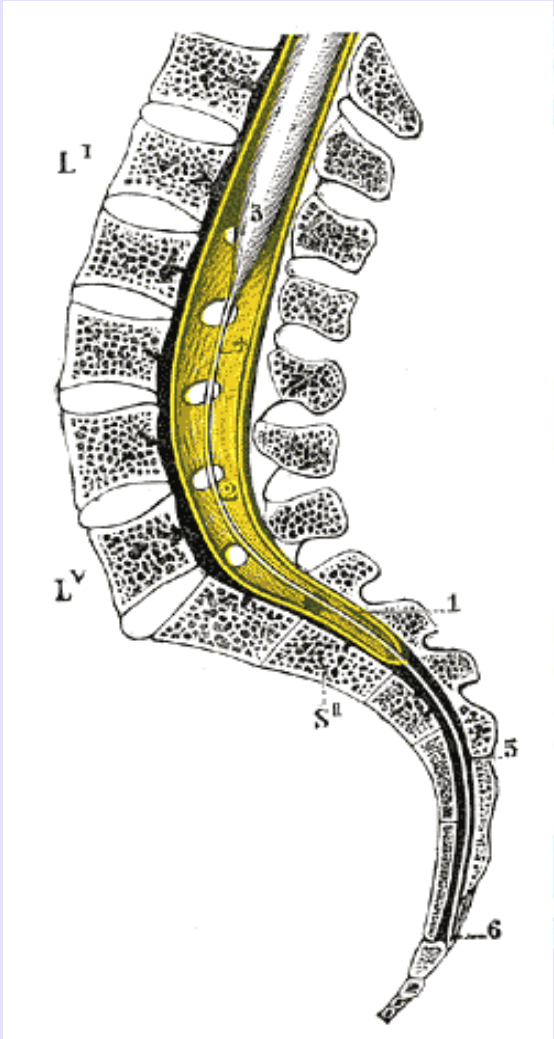
Subdural space

Subarachnoid space – between the pia mater and the arachnoid membrane

FILUM TERMINALE

- The **filum terminale** is a delicate filament, about 20 cm. in length, prolonged downward from the apex of the conus medullaris.
- It consists of two parts, an upper and a lower.
- The upper part, or ***filum terminale internum***, measures about 15 cm. in length and reaches as far as the lower border of the second sacral vertebra. It is contained within the tubular sheath of dura mater, and is surrounded by the nerves forming the cauda equina, from which it can be readily recognized by its bluish-white color.
- The lower part, or ***filum terminale externum***, is closely invested by, and is adherent to, the dura mater; it extends downward from the apex of the tubular sheath and is attached to the back of the first segment of the coccyx.

filum terminale(cont'd)



- Lower end of the medulla and the filum terminale.
  LI, LV. First and fifth lumbar vertebra.
  SII. Second sacral vertebra.
  1. Dura mater.
  2. Lower part of subarachnoid cavity.
  3. Lower extremity of medulla .
  4. Filum terminale internum.
  5. Filum terminale externum.
  6. Attachment of filum terminale to first segment of coccyx.

DENTICULATE LIGAMENT

- The pia mater of the spinal cord has 21 pairs of **denticulate ligaments** which attach it to the arachnoid and dura maters. Named for their tooth like appearance, the denticulate ligaments are traditionally believed to provide stability for the spinal cord against motion within the vertebral column.

PIA MATER

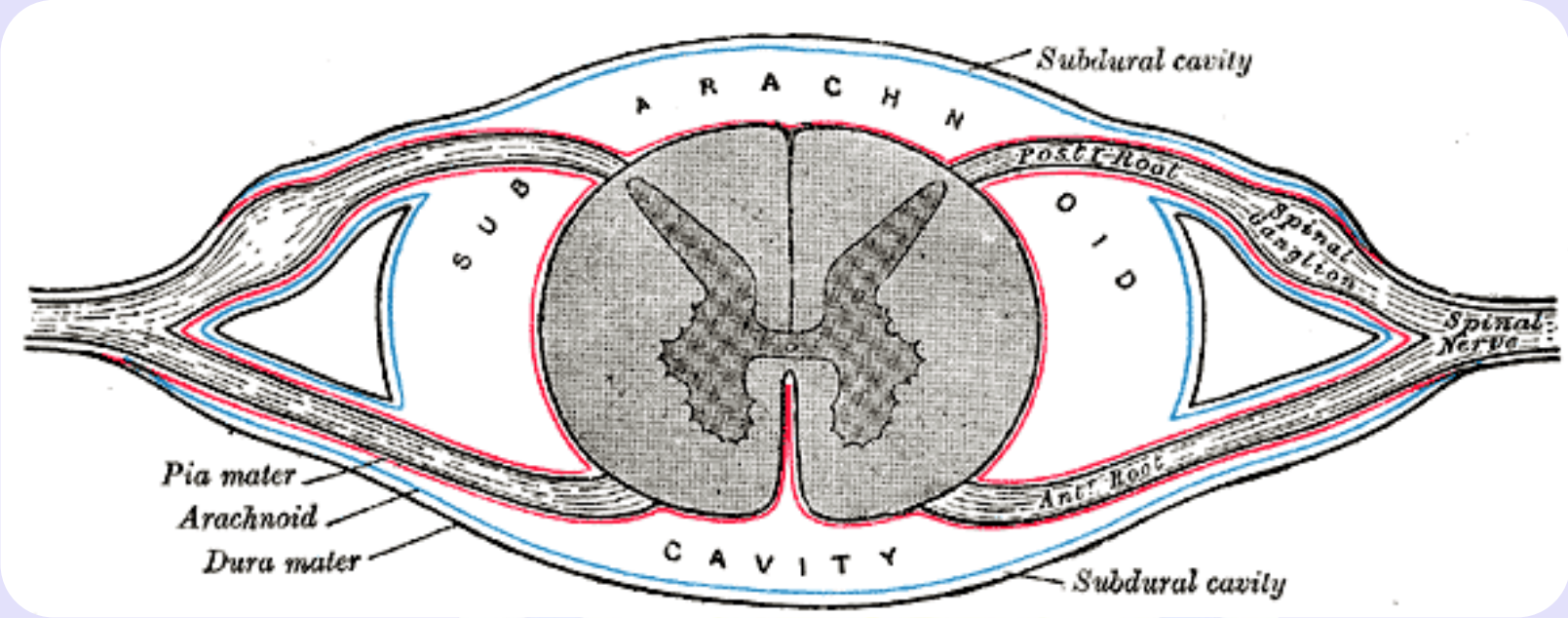
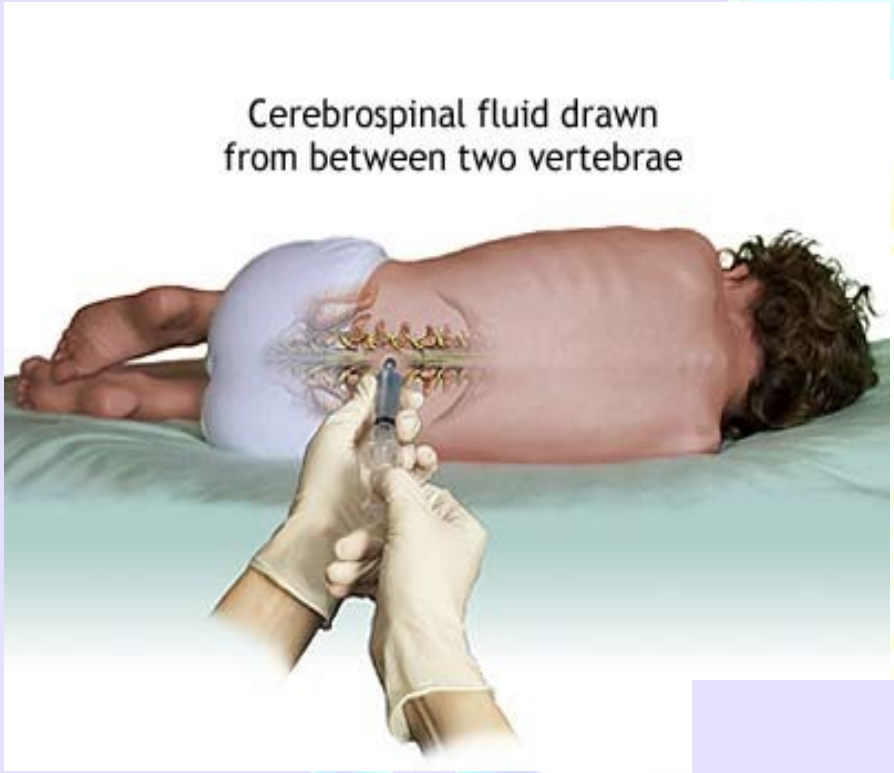


Diagram: transverse section of the medulla spinalis and its membranes. (At border, dura mater is black line, arachnoid mater is blue line, and pia mater is red line.)

MENINGITIS

- infection of the meninges (pia and arachnoid) due to organisms crossing the blood-brain barrier or choroid plexus
- wide range of organisms responsible
- 3500 – 4000 cases per year (UK)
- most common – viral meningitis
- less common but more severe – bacterial meningitis
- Notifiable disease (CDC)

DIAGNOSTIC TEST FOR MENINGITIS : LUMBAR PUNCTURE….



A lumbar puncture collects cerebrospinal fluid to check for the presence of disease or injury.

A spinal needle is inserted, usually between the 3rd and 4th lumbar vertebrae in the lower spine.

Permits the urgent distinction of bacterial meningitis from viral meningitis and examination of the CSF allows precise diagnosis.



THANK YOU